Register Number: 20101200 7

Name of the Candidate: BALAJI G.

3233

B.E. DEGREE EXAMINATION, 2023

(COMPUTER SCIENCE AND ENGINEERING)

(FOURTH SEMESTER)

(Common to CS, IT and AI & ML)

CSBS–401/ITBS–401/AIBS–401: DISCRETE MATHEMATICS

(For those who joined from 2018-19)

November] [Time : 3 Hours

Maximum : 75 Marks

PART A - (10 X 1 = 10)
Answer ALL Questions

		Marks	CO	PO	BTL
1.	What is mean by tautology?	1	CO1	1,2	K1
2.	Show that $\neg(P \rightarrow Q) \Leftrightarrow P \wedge \neg Q$	1	CO1	2,3	K2
3.	Define disjoint sets with an example	1	CO2	1,2	K1
4.	Define equivalence relation.	1	CO2	1,2	K1
5.	Define a distributive Lattice	1	CO3	2,3	K1
6.	State the isotonic property of a Lattice	1	CO3	1,2	K1
7.	Define a Monoid with an example	1	CO4	1,2	K1
8.	Define Hamming code	1	CO4	3,4	K2
9.	Define regular graph with an example	1	CO5	1,2	K1
10.	Define a circuit and a simple circuit of a graph.	1	CO5	1,2	K1

PART- B (5 x 3 = 15 Marks)
Answer either (a) or (b) in each Question

11.	(a) Prove that $\neg Q \wedge (P \rightarrow Q) \Rightarrow \neg P$	3	CO1	1,2	K2
	(OR)				
	(b) Obtain the Principal Conjunctive normal form of $(\neg P \rightarrow R) \wedge (Q \leftrightarrows P)$	3	CO1	2,3	K2
12.	(a) If A,B and C are any three sets, Prove that $(A - B) - C = (A - C) - (B - C)$	3	CO2	2,3	K2
	(OR)				

(b) If R is the relation on A = {1,2,3} such that $(a,b) \in R$ if and only if $a + b = even$, Find R^{-1}, $\bar{R}$ and R^2 using relational matrix M_R. | 3 | CO2 | 2,3 | K2 |

13. (a) Prove that D_{42} is a complemented lattice by finding the complements of all the elements | 3 | CO3 | 1,2 | K2 |

(OR)

(b) Show that $ab + abc + a'b + ab'c = b + ac$ in a Boolean algebra. | 3 | CO3 | 2,3 | K2 |

14. (a) Show that in a group G, | 3 | CO4 | 1,2 | K2 |

$(a * b)^{-1} = b^{-1} * a^{-1}$ *for all* $a, b \in G$

(OR)

(b) Prove that the set {1,3,7,9} is an abelian group under multiplication modulo 10 | 3 | CO4 | 3,4 | K2 |

15. (a) Explain Konigsberg bridge problem. Represent the problem by means of graph. Does the problem have a solution? | 3 | CO5 | 1,2 | K2 |

(OR)

(b) Define Bipartite graph. Draw $K_{2,3}$ and $K_{3,3}$ graphs. | 3 | CO5 | 2,3 | K2 |

PART- C (4 x10 = 40 Marks)
Answer either (a) or (b) in each Question

16. (a) Find whether the conclusion C: R follows from the premises $H_1: P \vee Q$; $H_2: \neg R \rightarrow \neg P$; $H_3: \neg(Q \wedge \neg R)$ using truth table technique. | 10 | CO1 | 2,3 | K3 |

(OR)

(b) Show that $\neg p$ is logically follows from the premises $(p \rightarrow q) \wedge (r \rightarrow s)$, $(q \rightarrow t) \wedge (s \rightarrow u)$, $\neg(t \wedge u)$ and $p \rightarrow r$ | 10 | CO1 | 3,4 | K3 |

17. (a) Show that the relation R "congruent modulo n" is an equivalence relation on the set of integers | 10 | CO2 | 2,3 | K3 |

(OR)

(b) Show that for any three sets $A \times (B \cap C) = (A \times B) \cap (A \times C)$ | 10 | CO2 | 1,3 | K3 |

18. (a) Find the disjunctive and conjunctive normal forms of the Boolean expression $f(x,y,z) = (yz + xz')(xy' + z)'$ | 10 | CO3 | 1,2 | K3 |

(OR)

(b) Simplify the Boolean function $f(a,b,c,d) = \Sigma(1,2,4,5,6,11,12,13,14,15)$ | 10 | CO3 | 2,3 | K3 |

19. (a) Show that $\{s_3,*\}$, where $s = \{1,2,3\}$ is a group under the operation of right composition of permutation | 10 | CO4 | 2,3 | K3 |

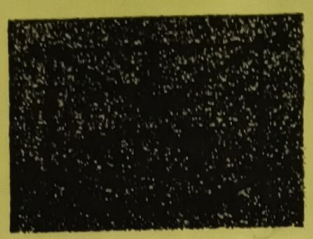
(OR)

(b) List the code words generated by the encoding function $e: B^3 \rightarrow B^6$ whose parity check matrix is | 10 | CO4 | 2,3 | K3 |

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

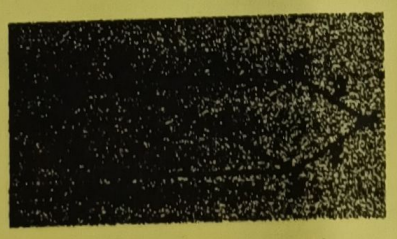
PART- D (1 x 10 = 10 Marks)
Answer either (a) or (b)

20. (a) Summarize all the paths of length 4 between D and E for the following graph using adjacency matrix. | 10 | CO5 | 2,3 | K4 |



(OR)

(b) Find the shortest path from the vertex A to the vertex F | 10 | CO5 | 2,3 | K4 |



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